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MOVABLE DEVICE, IMAGE PROJECTION APPARATUS, HEAD-UP DISPLAY, LASER HEADLAMP, HEAD-MOUNTED DISPLAY, OBJECT RECOGNITION APPARATUS, AND MOVING BODY

Abstract

A movable device includes a fixed frame, a movable portion, multiple drive beams to drive the movable portion. Each of the multiple drive beams has an end directly or indirectly coupled to the movable portion and another end coupled to the fixed frame. Each of the multiple drive beams includes: an upper drive beam and a lower drive beam separated from the upper drive beam in a normal direction of the fixed frame.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2024-027665, filed on Feb. 27, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a movable device, an image projection apparatus, a head-up display, a laser headlamp, a head-mounted display, an object recognition apparatus, and a moving body.

Related Art

[0003] A movable device such as a micro electro mechanical systems (MEMS) device manufactured by micromachining silicon or glass is known.

[0004] For example, in the related art, a displacement control actuator has been proposed that includes a piezoelectric element in which first main surfaces of at least two piezoelectric substrates having first and second main surfaces are bonded to each other by using direct bonding in order to obtain a large displacement amount.

[0005] However, in a movable device such as the displacement control actuator proposed in the related art, the rigidity of the deformable part is reduced in order to obtain a large displacement amount. The operation stability is lowered due to the reduced rigidity of the deformable part, and accordingly there is room for increasing the operation stability.

SUMMARY

[0006] According to an embodiment of the present disclosure, a movable device includes a fixed frame, a movable portion, and multiple drive beams to drive the movable portion. Each of the multiple drive beams has an end directly or indirectly coupled to the movable portion and another end coupled to the fixed frame. Each of the multiple drive beams includes an upper drive beam and a lower drive beam separated from the upper drive beam in a normal direction of the fixed frame.

[0007] According to an embodiment of the present disclosure, an image projection apparatus includes the movable device.

[0008] According to an embodiment of the present disclosure, a head-up display includes the movable device.

[0009] According to an embodiment of the present disclosure, a laser headlamp includes the movable device.

[0010] According to an embodiment of the present disclosure, a head-mounted display includes the movable device.

[0011] According to an embodiment of the present disclosure, an object recognition apparatus includes the movable device.

[0012] According to an embodiment of the present disclosure, a moving body includes the head-up display.

[0013] According to an embodiment of the present disclosure, a moving body includes the laser headlamp.

[0014] According to an embodiment of the present disclosure, a moving body includes the object recognition apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

[0016] FIG. 1 is a schematic top view of a movable device according to a first embodiment of the present disclosure;

[0017] FIG. 2 is a schematic cross-sectional view of the movable device taken along the line II-II in FIG. 1;

[0018] FIG. 3 is a schematic cross-sectional view of the movable device taken along the line III-III in FIG. 1;

[0019] FIG. 4 is a schematic bottom view of the movable device according to the first embodiment of the present disclosure;

[0020] FIG. 5 is a schematic cross-sectional view of the movable device taken along the line V-V in FIG. 1;

[0021] FIG. 6 is a first schematic perspective view of a movable portion oscillated by a second upper drive portion in the movable device according to the first embodiment of the present disclosure;

[0022] FIG. 7 is a second schematic perspective view of the movable portion oscillated by the second upper drive portion in the movable device according to the first embodiment of the present disclosure;

[0023] FIG. 8 is a third schematic perspective view of the movable portion oscillated by the second upper drive portion in the movable device according to the first embodiment of the present disclosure;

[0024] FIG. 9 is a fourth schematic perspective view of the movable portion oscillated by the second upper drive portion in the movable device according to the first embodiment of the present disclosure;

[0025] FIG. 10 is a first diagram illustrating a waveform of a drive voltage applied to a drive portion group of the movable device according to the first embodiment of the present disclosure;

[0026] FIG. 11 is a second diagram illustrating a waveform of a drive voltage applied to a drive portion group of the movable device according to the first embodiment of the present disclosure;

[0027] FIG. 12 is a diagram illustrating a waveform the waveform of the drive voltage in FIG. 10 and the waveform of the drive voltage in FIG. 11 are superimposed;

[0028] FIG. 13 is a schematic top view of a movable device according to a second embodiment of the present disclosure;

[0029] FIG. 14 is a schematic cross-sectional view of the movable device taken along the line XIV-XIV in FIG. 13;

[0030] FIG. 15 is a schematic cross-sectional view of the movable device taken along the line XV-XV in FIG. 13;

[0031] FIG. 16 is a schematic bottom view of the movable device according to the second embodiment of the present disclosure;

[0032] FIG. 17 is a schematic cross-sectional view of the movable device taken along the line XVII-XVII in FIG. 13;

[0033] FIG. 18 is a schematic top view of a movable device according to a third embodiment of the present disclosure;

[0034] FIG. 19 is a schematic top view of the movable device taken along the line XIV-XIV in FIG. 20;

[0035] FIG. 20 is a schematic cross-sectional view of the movable device taken along the line XX-XX in FIG. 18;

[0036] FIG. 21 is a schematic cross-sectional view of the movable device taken along the line XXI-

XXI in FIG. 18;

[0037] FIG. 22 is a schematic bottom view of the movable device according to the third embodiment of the present disclosure;

[0038] FIG. 23 is a schematic cross-sectional view of the movable device taken along the line XXIII-XXIII in FIG. 18;

[0039] FIG. 24 is a schematic top view of a movable device according to a fourth embodiment of the present disclosure;

[0040] FIG. 25 is a schematic cross-sectional view of the movable device taken along the line XXV-XXV in FIG. 24;

[0041] FIG. 26 is a schematic cross-sectional view of the movable device taken along the line XXVI-XXVI in FIG. 24;

[0042] FIG. 27 is a schematic bottom view of the movable device according to the fourth embodiment of the present disclosure;

[0043] FIG. 28 is a schematic cross-sectional view of the movable device taken along the line XXVIII-XXVIII in FIG. 24;

[0044] FIG. 29 is a schematic top view of a movable device according to a fifth embodiment of the present disclosure;

[0045] FIG. 30 is a schematic cross-sectional view of the movable device taken along the line XXX-XXX in FIG. 29;

[0046] FIG. 31 is a schematic cross-sectional view of the movable device taken along the line XXXI-XXXI in FIG. 29;

[0047] FIG. 32 is a schematic bottom view of the movable device according to the fifth embodiment of the present disclosure;

[0048] FIG. 33 is a schematic cross-sectional view of the movable device taken along the line XXXIII-XXXIII in FIG. 29;

[0049] FIG. 34 is a schematic top view of a movable device according to a sixth embodiment of the present disclosure;

[0050] FIG. 35 is a schematic cross-sectional view of the movable device taken along the line XXXV-XXXV in FIG. 34;

[0051] FIG. 36 is a schematic cross-sectional view of the movable device taken along the line XXXVI-XXXVI in FIG. 34;

[0052] FIG. 37 is a schematic bottom view of the movable device according to the sixth embodiment of the present disclosure;

[0053] FIG. 38 is a schematic cross-sectional view of the movable device taken along the line XXXVIII-XXXVIII in FIG. 34;

[0054] FIG. 39 is a schematic diagram illustrating an example of a light scan system;

[0055] FIG. 40 is a diagram illustrating a hardware configuration of a light scan system;

[0056] FIG. 41 is a functional block diagram illustrating an example of a control device;

[0057] FIG. 42 is a flowchart of an example of a process related to a light scan system;

[0058] FIG. 43 is a schematic diagram illustrating an example of an automobile including a head-up display apparatus;

[0059] FIG. 44 is a schematic diagram illustrating an example of a head-up display apparatus;

[0060] FIG. 45 is a schematic diagram illustrating an example of an image forming apparatus including an optical writing device;

[0061] FIG. 46 is a schematic diagram illustrating an example of an optical writing device;

[0062] FIG. 47 is a schematic diagram illustrating an example of an automobile including a laser imaging detection and ranging apparatus;

[0063] FIG. 48 is a schematic diagram illustrating an example of a laser imaging detection and ranging apparatus;

[0064] FIG. 49 is a schematic diagram illustrating an example of a configuration of a laser

headlamp;

[0065] FIG. **50** is a schematic perspective view of an example of a configuration of a head-mounted display.

[0066] FIG. **51** is a schematic diagram illustrating an example of a portion of a configuration of a head-mounted display;

[0067] FIG. **52** is a first schematic diagram illustrating an example of a pupil-or-cornea position detection apparatus; and

[0068] FIG. **53** is a second schematic diagram illustrating an example of a pupil-or-cornea position detection apparatus.

[0069] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

[0070] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

[0071] Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0072] According to an embodiment of the present disclosure, a movable device that can obtain a large displacement amount while increasing a displacement stability can be provided.

[0073] A movable device, an image projection apparatus, a head-up display, a laser headlamp, a head-mounted display, an object recognition apparatus, and a moving body according to embodiments of the present disclosure will be described in detail with reference to the drawings. However, the following embodiments are intended to describe the movable device, the image projection apparatus, the head-up display, the laser headlamp, the head-mounted display, the object recognition apparatus, and the moving body according to some embodiments of the present disclosure, and the scope of the present disclosure is not limited to the following embodiments.

[0074] The dimensions, materials, shapes, and relative positions of the components described in embodiments of the present disclosure are not intended to limit the scope of the embodiments of the present disclosure to only those of the components, and are merely examples of explanation, unless otherwise specified. The relative positions or the size of the elements illustrated in the drawings may be exaggerated for purpose of clear description. In the following description, common or corresponding elements are denoted by the same or similar reference signs, and redundant description is appropriately simplified or omitted.

[0075] In the following, the arrangement and configuration of the respective parts will be described using an XYZ orthogonal coordinate system for the sake of easy understanding of the description. The three axes in the XYZ orthogonal coordinate system are orthogonal to each other. In the XYZ orthogonal coordinate system, a direction in which the X-axis extends is referred to as an “X-direction”, a direction in which the Y-axis extends is referred to as a “Y-direction”, and a direction in which the Z-axis extends is referred to as a “Z-direction”. The direction in which the arrow indicating the X-axis is directed is referred to as +X-direction, and the direction opposite to the +X-direction is referred to as −X-direction. The direction in which the arrow indicating the Y-axis is directed is referred to as +Y-direction, and the direction opposite to the +Y-direction is referred to as −Y-direction. The direction in which the arrow indicating the Z-axis is directed is referred to as +Z-direction, and the direction opposite to the +Z-direction is referred to as −Z-direction.

[0076] In embodiments of the present disclosure, the +Z-direction is referred to as “upper” and the

-Z-direction is referred to as “lower”. Viewing an object from the +Z-direction is referred to as top view. However, these directional representations merely describe the relation of relative positions, orientations, and directions, and may not be consistent with the relation in use. These directions are independent of the direction of gravity. For example, in an embodiment of the present disclosure, an upper drive portion may not be disposed on the upper side in the gravity direction, but may be disposed on the relatively opposite side to a lower drive portion.

[0077] The term “disposed” is not only limited to the case of direct contact, but also includes the case of indirect disposition, for example, with another member interposed. In the present specification, the term “orthogonal” may include an error within the range of $\pm 10^\circ$ with respect to 90° . Further, in the present specification, the term “parallel” may include an error within the range of $\pm 10^\circ$ with respect to 0° .

First Embodiment

Structure of Movable Device According to First Embodiment

[0078] A movable device according to a first embodiment of the present disclosure will be described with reference to FIGS. 1 to 5. FIG. 1 is a schematic top view of an example of the movable device **13** according to the first embodiment of the present disclosure. FIG. 2 is a schematic cross-sectional view of the movable device **13** taken along the line II-II in FIG. 1. FIG. 3 is a schematic cross-sectional view of the movable device **13** taken along the line III-III in FIG. 1. FIG. 4 is a schematic bottom view of an example of the movable device **13** according to the first embodiment of the present disclosure. FIG. 5 is a schematic cross-sectional view of the movable device **13** taken along the line V-V in FIG. 1.

[0079] The movable device **13** includes a movable portion **101**, a drive beam **112**, and a fixed frame **140**. The drive beam **112** has one end **135** (see FIG. 1) indirectly coupled to the movable portion **101** via a support portion **120** and drives the movable portion **101**. The other end **136** (see FIG. 4) of the drive beam **112** coupled to the fixed frame **140**. In the example illustrated in FIGS. 1 to 5, the movable device **13** includes a support portion **120** and electrode connecting portions **150**.

[0080] The drive beam **112** includes multiple drive beams each having a drive portion **131**. As illustrated in FIGS. 2 and 3, the drive beam **112** includes multiple upper drive beams **112u** and multiple lower drive beams **112v** arranged apart from the multiple upper drive beams **112u** in the normal direction of the fixed frame **140**. Each of the multiple upper drive beams **112u** includes a first silicon on insulator (SOI) substrate **361** and an upper drive portion **131u**. Each of the multiple lower drive beams **112v** includes a second SOI substrate **362** and a lower drive portion **131v**.

[0081] The normal direction of the fixed frame **140** is a direction parallel to the normal of a virtual plane including the outer edge of the fixed frame **140** in a top view. In another aspect, the normal direction of the fixed frame **140** is a direction orthogonal to the first axis **E1** and the second axis **E2**. In still another aspect, the normal direction of the fixed frame **140** is a direction parallel to the normal of the movable portion **101** in a state where the movable portion **101** is not driven by the drive beam **112**. In the examples illustrated in FIGS. 1 to 5, the normal of the fixed frame **140** is along the Z-axis.

[0082] In the example illustrated in FIG. 1, the movable portion **101** is driven by the drive beam **112** so that the movable portion is displaced so as to oscillate (rotationally vibrate) about the second axis **E2** along the Y-axis as a rotation axis. The drive beam **112** includes a first upper drive beam **112u-1** arranged at one side (+Y-direction) of the movable portion **101** in the direction along the second axis **E2**, and a second upper drive beam **112u-2** arranged at the other side (-Y-direction) of the movable portion **101**. The drive portion **131** includes a first upper drive portion **131u-1** of the first upper drive beam **112u-1** and a second upper drive portion **131u-2** of the second upper drive beam **112u-2**. The one end **135** includes a first end **135-1** at which the first upper drive beam **112u-1** is coupled to the movable portion **101**, and a second end **135-2** at which the second upper drive beam **112u-2** is coupled to the movable portion **101**.

[0083] In the example illustrated in FIG. 1, in order to indicate that the drive beam **112** includes the

first upper drive beam **112u-1**, the reference sign of the first upper drive beam **112u-1** is written together with the reference sign of the drive beam **112** in parentheses. In addition, in order to indicate that the drive beam **112** includes the second upper drive beam **112u-2**, the reference sign of the second upper drive beam **112u-2** is written together with the reference sign of the drive beam **112** in parentheses. In addition, in order to indicate that the drive portion **131** includes the first upper drive portion **131u-1**, the reference sign of the first upper drive portion **131u-1** is written together with the reference sign of the drive portion **131** in parentheses. In addition, in order to indicate that the drive portion **131** includes the second upper drive portion **131u-2**, the reference sign of the second upper drive portion **131u-2** is written together with the reference sign of the drive portion **131** in parentheses. In the following drawings other than FIG. 1, reference signs may be written together for the same purpose.

[0084] As illustrated in FIG. 2, the drive beam **112** includes multiple upper drive beams **112u** and multiple lower drive beams **112v**. The upper drive beam **112u** includes a first upper drive beam **112u-1** and a second upper drive beam **112u-2**. The lower drive beam **112v** includes a first lower drive beam **112v-1** and a second lower drive beam **112v-2**. The drive portion **131** includes an upper drive portion **131u** disposed at the upper drive beam **112u** and a lower drive portion **131v** disposed at the lower drive beam **112v**. The upper drive portion **131u** includes a first upper drive portion **131u-1** disposed at the first upper drive beam **112u-1** and a second upper drive portion **131u-2** disposed at the second upper drive beam **112u-2**. The lower drive portion **131v** includes a first lower drive portion **131v-1** disposed at the first lower drive beam **112v-1** and a second lower drive portion **131v-2** disposed at the second lower drive beam **112v-2**. In the example illustrated in FIG. 4, the other end **136** includes a first other end **136-1** to which the first lower drive beam **112v-1** is coupled to the fixed frame **140**, and a second other end **136-2** at which the second lower drive beam **112v-2** is coupled to the fixed frame **140**.

[0085] In the example illustrated in FIGS. 1 and 2, the first upper drive portion **131u-1** includes a first upper drive portion **131a**, a first upper drive portion **131b**, a first upper drive portion **131c**, and a first upper drive portion **131d**. The second upper drive portion **131u-2** includes a second upper drive portion **132a**, a second upper drive portion **132b**, a second upper drive portion **132c**, and a second upper drive portion **132d**. In the example illustrated in FIG. 2, the first lower drive portion **131v-1** includes a first lower drive portion **133a**, a first lower drive portion **133b**, a first lower drive portion **133c**, and a first lower drive portion **133d**. The second lower drive portion **131v-2** includes a second lower drive portion **134a**, a second lower drive portion **134b**, a second lower drive portion **134c**, and a second lower drive portion **134d**.

[0086] In the movable device, for example, when a large drive voltage is applied to the drive beam in order to displace the movable portion by a large displacement amount, the drive portion such as the piezoelectric film may not withstand the drive voltage, and the drive beam may be broken. In addition, when the rigidity of the structure supporting the movable portion is reduced in order to displace the movable portion by a large displacement amount, the resonance frequency of the movable device is lowered. Accordingly, the displacement stability of the movable portion may be impaired.

[0087] In the movable device **13** according to an embodiment of the present disclosure, the drive beam **112** includes an upper drive beam **112u** and a lower drive beam **112v**, and displaces the movable portion **101** by a displacement amount obtained by combining the displacement by the upper drive beam **112u** and the displacement by the lower drive beam **112v**. Accordingly, the movable portion **101** can be displaced by a large displacement amount without applying a large drive voltage to the drive beam **112**. Further, since the structure that supports the movable portion **101** is formed so as not to lower the rigidity of the structure, the displacement stability of the movable portion **101** is not impaired. As described above, in the present embodiment, the movable device **13** that can obtain a large displacement amount while increasing the displacement stability can be provided.

[0088] In the movable device **13**, the upper drive beam **112u** overlaps the lower drive beam **112v** as viewed from the normal direction of the fixed frame **140**. As viewed from the normal direction of the fixed frame **140**, the upper drive beam **112u** overlaps the lower drive beam **112v**. Thus, the area of the movable device **13** in a virtual plane orthogonal to the normal of the fixed frame **140** is reduced, and the movable device **13** can be reduced in size, as compared with the case where the upper drive beam **112u** does not overlap the lower drive beam **112v**. In the overlap between the upper drive beam **112u** and the lower drive beam **112v**, at least a portion of the upper drive beam **112u** may overlap the lower drive beam **112v** as viewed from the normal direction of the fixed frame **140**. When substantially the entire upper drive beam **112u** overlaps the lower drive beam **112v** as viewed from the normal direction of the fixed frame **140**, the area of the movable device **13** in the virtual plane orthogonal to the normal of the fixed frame **140** is minimized, and the movable device **13** can be reduced in size to the minimum.

[0089] In the movable device **13**, the upper drive beam **112u** is coupled to the lower drive beam **112v** by an adhesive. In the example illustrated in FIG. 5, the upper drive beam **112u** is coupled to the lower drive beam **112v** by an adhesive at the inter-substrate coupling portion **160**. The adhesive may be appropriately selected depending on the materials of the upper drive beam **112u** and the lower drive beam **112v**.

[0090] In the case where the drive portion such as a piezoelectric film is formed in a multilayer in order to displace the movable portion by a large displacement amount, the film is formed in a multilayer by a film forming process such as physical vapor deposition, sputtering, or chemical vapor deposition. In the film forming process, in order to form a film with high accuracy, it is necessary to optimize the manufacturing process or various parameters for each manufacturing process. Thus, the difficulty of the method for manufacturing the movable device may be increased. As the difficulty level of the method for manufacturing increases, the yield of the movable device may decrease.

[0091] In the movable device **13**, since the upper drive beam **112u** and the lower drive beam **112v** are mechanically coupled by an adhesive, the upper drive beam **112u** and the lower drive beam **112v** can be coupled more easily than in the film forming process. Thus, the movable device **13** that can obtain a large displacement amount while increasing the displacement stability can be easily manufactured. In addition, the yield of the movable device **13** can be increased by simplifying the manufacturing.

[0092] In the movable device **13**, the movable portion **101** is disposed so as to be spaced from either the upper drive beam **112u** or the lower drive beam **112v** in the normal direction of the fixed frame **140**. In the movable device **13**, the movable portion **101** is disposed so as to be spaced from either the upper drive beam **112u** or the lower drive beam **112v** in the top view. Accordingly, the movable device **13** can be reduced in size as compared with the case where the movable portion **101** is disposed so as not to overlap with either the upper drive beam **112u** or the lower drive beam **112v** in a top view. In addition, the movable device **13** is reduced in size so that the resonance frequency in the oscillation of the movable portion **101** about the second axis E2 as the rotation axis can become higher.

[0093] The one end **135** of the drive beam **112** may not be coupled to the movable portion **101** indirectly via the support portion **120**, but may be coupled directly to the movable portion **101**.

[0094] A configuration of the movable device **13** will be described in detail below.

[0095] The movable portion **101** includes a movable portion base **102** and a mirror surface **14** disposed above the movable portion base **102**. The mirror surface **14** reflects a light beam incident on the mirror surface **14**. The movable device **13** can oscillate the movable portion **101** about the first axis E1 as the rotation axis and oscillate the movable portion **101** about the second axis E2 as the rotation axis. The inclination of the mirror surface **14** changes in accordance with the oscillation of the movable portion **101**. The movable device **13** changes the inclination of the mirror surface **14** by oscillating the movable portion so that the movable device can deflect the light beam incident

on the mirror surface **14** and scan an object with the light beam in the direction along the first axis **E1** and the second axis **E2**.

[0096] The support portion **120** includes a first torsion bar spring **111a**, a second torsion bar spring **111b**, a third drive portion **112a**, and a fourth drive portion **112b**. The support portion **120** supports the movable portion **101** and oscillates the movable portion **101** about the first axis **E1** as a rotation axis by the third drive portion **112a** and the fourth drive portion **112b**. The support portion **120** includes, for example, a silicon support layer **361a**, a silicon oxide layer **361b**, and a silicon active layer **361c**. The support portion **120** includes a frame having a substantially rectangular outer edge shape in a top view. The frame of the support portion **120** surrounds the movable portion **101**, the third drive portion **112a**, and the fourth drive portion **112b** in a top view.

[0097] The first torsion bar spring **111a** and the second torsion bar spring **111b** are torsion beams. In the example illustrated in FIG. **1**, the first torsion bar spring **111a** is disposed in the $-X$ -direction of the movable portion **101**. One end of the first torsion bar spring **111a** is coupled to the movable portion **101**. The other end of the first torsion bar spring **111a** is coupled to a base member on which the third drive portion **112a** is disposed via the drive portion coupling portion **113a**. The second torsion bar spring **111b** is disposed at the movable portion **101** in the $+X$ -direction. One end of the second torsion bar spring **111b** is coupled to the movable portion **101**. The other end of the second torsion bar spring **111b** is coupled to the base member on which the fourth drive portion **112b** is disposed via the drive portion coupling portion **113b**. The third drive portion **112a** drives the movable portion **101** via the first torsion bar spring **111a**. The fourth drive portion **112b** drives the movable portion **101** via the second torsion bar spring **111b**.

[0098] The fixed frame **140** is a frame member having a substantially rectangular outer edge shape in a top view. The movable portion **101**, the drive beam **112**, and the support portion **120** are disposed inside the fixed frame **140**. The electrode connecting portions **150** are disposed at an $+X$ -side end on the $+Z$ -side surface of the fixed frame **140**. The electrode connecting portions **150** electrically connect the first drive portion **131-1**, the second drive portion **131-2**, the third drive portion **112a**, and the fourth drive portion **112b** to a control device that controls the driving of the movable device **13**. The drive voltage from the control device is applied to the first drive portion **131-1**, the second drive portion **131-2**, the third drive portion **112a**, and the fourth drive portion **112b** via the electrode connecting portions **150**.

[0099] As illustrated in FIG. **3**, the movable device **13** is formed by bonding two SOI substrates, i.e., the first SOI substrate **361** and the second SOI substrate **362**, with a silicon oxide layer **363** put therebetween. An upper electrode **201**, a lower electrode **203**, and a piezoelectric portion **202** put between the lower electrode **203** and the upper electrode **201** are formed on the surfaces of the first SOI substrate **361** and the second SOI substrate **362**. Each drive portion included in the drive portion **131** includes a piezoelectric portion **202** put between a lower electrode **203** and an upper electrode **201**.

[0100] Each of the first SOI substrate **361** and the second SOI substrate **362** is a substrate in which a silicon oxide layer is disposed above a first silicon layer made of single-crystal silicon (Si), and a second silicon layer made of single-crystal silicon is further disposed above the silicon oxide layer. In the following description, the first silicon layer is referred to as a silicon support layer **361a** and a silicon support layer **362a**, and the second silicon layer is referred to as a silicon active layer **361c** and a silicon active layer **362c**.

[0101] The first SOI substrate **361** includes the silicon support layer **361a**, the silicon oxide layer **361b**, and the silicon active layer **361c**. The second SOI substrate **362** includes the silicon support layer **362a**, the silicon oxide layer **362b**, and the silicon active layer **362c**. The first SOI substrate **361** and the second SOI substrate **362** are formed by etching. The mirror surface **14**, the first drive portion **131-1**, the second drive portion **131-2**, the third drive portion **112a**, the fourth drive portion **112b**, and the electrode connecting portions **150** are formed above each of the first SOI substrate **361** and the second SOI substrate **362**, so that these portions can be integrally formed in one body.

The above-described portions may be formed after the formation of the first SOI substrate **361** and the second SOI substrate **362**, or may be formed during the formation of the first SOI substrate **361** and the second SOI substrate **362**. The method for bonding the two SOI substrates, i.e., the first SOI substrate **361** and the second SOI substrate **362**, is not limited to bonding with an adhesive, and may be joining. In the case of bonding, the silicon oxide layer **363** serves as an adhesive.

[0102] Since the silicon active layer **361c** has a small thickness in the Z-direction relative to the X-direction and the Y-direction, a member including only the silicon active layer **361c** has a function as an elastic portion having elasticity. The first SOI substrate **361** and the second SOI substrate **362** may not be planar, but may have curvature. The base member used for forming the movable device **13** is not limited to the SOI substrate as long as the base member can be integrally formed as one body by etching and can be partially given elasticity.

[0103] The details of the main scanning structure for performing oscillation with the first axis **E1** as the rotation axis will be described. The movable portion base **102** includes, for example, a silicon active layer **361c**. The mirror surface **14** includes a thin metal film containing, for example, aluminum, gold, or silver. In the movable portion **101** illustrated in FIG. 4, a rib **103** for reinforcing the mirror portion is formed at the -Z-side surface of the movable portion base **102**. The rib **103** includes, for example, a silicon support layer **361a**, and a silicon oxide layer **361b**. The rib **103** can reduce the distortion of the mirror surface **14** caused when the movable portion **101** is oscillated.

[0104] The fourth driving portion **112b** illustrated in FIG. 3 is formed by laminating an upper electrode **201**, a piezoelectric portion **202**, and a lower electrode **203** in this order above the +Z-side surface of a silicon active layer **361c** that is an elastic portion. The upper electrode **201** and the lower electrode **203** are made of, for example, gold (Au) or platinum (Pt). The piezoelectric portion **202** is made of, for example, lead zirconate titanate (PZT) that is a piezoelectric material. The configuration of the third drive portion **112a** is also the same as that of the fourth drive portion **112b**.

[0105] As illustrated in FIG. 4, the first torsion bar spring **111a** and the second torsion bar spring **111b** are members extending in the direction along the first axis **E1**. The first torsion bar spring **111a** and the second torsion bar spring **111b** include the silicon active layer **361c** of the first SOI substrate **361**. Each of the third drive portion **112a** and the fourth drive portion **112b** has a unimorph structure and is deformed only in one direction by the application of a drive voltage. However, the movable device **13** can reciprocally oscillate the movable portion **101** supported by the support portion **120** by the action of the respective springs of the first torsion bar spring **111a** and the second torsion bar spring **111b**. The oscillation of the movable portion **101** about the first axis **E1** as the rotation axis is resonant oscillation. The resonance frequency is designed using information such as the moment of inertia of the movable portion **101** or the rigidity of the first torsion bar spring **111a** and the second torsion bar spring **111b**.

[0106] As illustrated in FIG. 4, the torsion center axis **C0** of the first torsion bar spring **111a** and the second torsion bar spring **111b** is offset from the first axis **E1** passing through the center of the movable portion **101**. Due to this offset, the movable device **13** changes the Z-directional displacement of the tip in the +Y-direction of the base member at which the third drive portion **112a** and the fourth drive portion **112b** are disposed into a rotational force (moment) to oscillate the movable portion **101**. In the example illustrated in FIG. 4, the torsion center axis **C0** is the torsion center axis of the first torsion bar spring **111a** and the torsion center axis of the second torsion bar spring **111b**. The offset ΔS represents an offset amount of the torsion center axis **C0** with respect to the first axis **E1**.

[0107] The base member at which the third drive portion **112a** and the fourth drive portion **112b** are disposed has a free end at the tip thereof, and a large displacement can be obtained without preventing the piezoelectric driving force. Further, the bending mode resonance of the base member at which the third drive portion **112a** and the fourth drive portion **112b** are disposed is set in the vicinity of the torsional mode resonance of each of the first torsion bar spring **111a** and the

second torsion bar spring **111b** so that the rotation angle of the movable portion **101** can be greatly increased, and a large displacement can be obtained even when a low drive voltage is applied. [0108] The structure for performing oscillation with the second axis **E2** as a rotation axis will be described in detail. The oscillation of the movable portion **101** about the first axis **E1** as the rotation axis is resonant oscillation. Thus, the waveform of the drive voltage becomes a sine wave. In many cases, the movable portion **101** performs a constant-speed oscillation with the second axis **E2** as the rotation axis. In addition, a sawtooth waveform is often used for the waveform of the drive voltage. In order to perform the constant-speed oscillation by the drive voltage having the sawtooth waveform, it is preferable to perform the non-resonant oscillation. In the non-resonant oscillation, a large displacement as in the resonant oscillation is difficult to obtain. In the oscillation with the second axis **E2** as the rotation axis, it is preferable that the drive beam is deformed in both directions. As described above, the structure of the drive beam is a meander structure in which multiple drive beams are folded back. The meander structure allows the drive beam to be deformed in both directions by performing oscillation of two channels, i.e., the A channel and the B channel. Further, the displacement of the movable portion **101** can be increased by accumulating the deformations of the multiple drive beams.

[0109] In the example illustrated in FIGS. **1** to **3**, the upper drive beam **112u** is formed with the first SOI substrate **361**. The second SOI substrate **362** forms the lower drive beam **112v**. The upper drive portion **131u** is formed in the silicon active layer **361c** of the first SOI substrate **361**. The lower drive portion **131v** is formed in the silicon active layer **362c** of the second SOI substrate **362**. Each of the upper drive beam **112u** and the lower drive beam **112v** has a meander structure.

[0110] A first end **135-1** at which a first upper drive beam **112u-1** is coupled to a movable portion **101**, a second end **135-2** at which a second upper drive beam **112u-2** is coupled to the movable portion **101**, a coupling portion at which the first upper drive beam **112u-1** is coupled to the first lower drive beam **112v-1**, a coupling portion at which the second upper drive beam **112u-2** is coupled to the second lower drive beam **112v-2**, a coupling portion at which the first lower drive beam **112v-1** is coupled to a fixed frame **140**, and a coupling portion at which the second lower drive beam **112v-2** is coupled to the fixed frame **140** are point symmetric with respect to the center of the mirror surface **14**.

[0111] As illustrated in FIG. **2**, the first upper drive portion **131u-1** and the second upper drive portion **131u-2** are formed by laminating a lower electrode **203**, a piezoelectric portion **202**, and an upper electrode **201** in this order above the surface of the silicon active layer **361c**, which is an elastic portion, in the +Z-direction. The first lower drive portion **131v-1** and the second lower drive portion **131v-2** are formed by laminating a lower electrode **203**, a piezoelectric portion **202**, and an upper electrode **201** in this order above the surface of the silicon active layer **362c**, which is an elastic portion, in the +Z-direction. The upper electrode **201** and the lower electrode **203** are made of, for example, gold (Au) or platinum (Pt). The piezoelectric portion **202** is made of, for example, lead zirconate titanate (PZT) that is a piezoelectric material.

[0112] In the example illustrated in FIGS. **1** and **3**, the fixed frame **140** is formed by bonding two SOI substrates, i.e., the first SOI substrate **361** and the second SOI substrate **362**, with the silicon oxide layer **363** put therebetween.

[0113] In the example illustrated in FIG. **5**, the upper drive beam **112u** is electrically connected to the lower drive beam **112v** via wire bonding **161**. In another aspect, in the inter-substrate coupling portion **160**, electrical wiring is formed between the first SOI substrate **361** and the second SOI substrate **362** via wire bonding **161** in order to apply a drive voltage to the piezoelectric portion **202**. Since the upper drive beam **112u** and the lower drive beam **112v** are electrically connected via the wire bonding **161**, the wiring between the first SOI substrate **361** and the second SOI substrate **362** that are different substrates can be performed. Since the wiring between the first SOI substrate **361** and the second SOI substrate **362** is performed, a drive voltage can be applied to each of the upper drive beam **112u** and the lower drive beam **112v**. The upper electrode **201** and the lower

electrode **203** may be directly connected to the electrode connecting portions **150**, or may be indirectly connected to the electrode connecting portions **150** by connecting the electrodes to each other.

[0114] The piezoelectric portion **202** is not limited to the configuration of being formed only on the +Z-side surface of the silicon active layer **361c**, which is an elastic portion. The piezoelectric portion **202** may be disposed on the -Z-side surface of the silicon active layer **361c** or may be disposed on both the +Z-side surface and the -Z-side surface of the silicon active layer **361c**. Each piezoelectric portion **202** may be formed of a multilayer piezoelectric film instead of a single layer.

[0115] The shapes of the components are not limited to the shapes illustrated in FIGS. **1** to **5** as long as the movable portion **101** can oscillate about the first axis E1 and the second axis E2. For example, the shapes of the first torsion bar spring **111a**, the second torsion bar spring **111b**, the upper drive beam **112u**, and the lower drive beam **112v** may have curvatures.

[0116] An insulating layer made of a silicon oxide film may be formed on at least one of the +Z-side surfaces of the upper electrodes **201** of the upper drive portion **131u**, the lower drive portion **131v**, the third drive portion **112a**, and the fourth drive portion **112b**, the +Z-side surface of the support portion **120**, and the +Z-side surface of the fixed frame **140**. In the case of forming an insulating layer, electrode wiring may be disposed above the insulating layer, and only a connection spot at which the upper electrode **201** or the lower electrode **203** is connected to the electrode wiring may be formed by partially removing the insulating layer as an opening or not forming the insulating layer. Accordingly, the degree of freedom in designing the upper drive portion **131u**, the lower drive portion **131v**, the third drive portion **112a**, the fourth drive portion **112b**, and the electrode wiring can be increased, and can reduce short circuits due to contact between electrodes. The silicon oxide film also functions as an anti-reflection material.

Oscillation of Movable Portion in Movable Device

[0117] The oscillation of the movable portion **101** by the upper drive portion **131u**, the lower drive portion **131v**, the third drive portion **112a**, and the fourth drive portion **112b** in the movable device **13** will be described in detail.

[0118] In the movable device **13**, in order to oscillate the movable portion **101**, a positive or negative drive voltage is applied to the piezoelectric portions **202** of the upper drive portion **131u**, the lower drive portion **131v**, the third drive portion **112a**, and the fourth drive portion **112b** with respect to the polarization direction. The drive voltage causes the piezoelectric portion **202** to deform (e.g., expand or contract) in proportion to the potential of the applied drive voltage, and what is called an inverse piezoelectric effect is obtained. Each of the upper drive portion **131u**, the lower drive portion **131v**, the third drive portion **112a**, and the fourth drive portion **112b** oscillates the movable portion **101** by utilizing the inverse piezoelectric effect.

[0119] In the following description, an angle formed by the mirror surface **14** of the movable portion **101** and the XY-plane when the mirror surface **14** is tilted in the +Z-direction or the -Z-direction with respect to the XY-plane is referred to as a deflection angle. A direction in which the +X-side of the movable portion **101** is tilted in the +Z-direction is defined as a positive deflection angle, and a direction in which the +X-side of the movable portion **101** is tilted in the -Z-direction is defined as a negative deflection angle.

Oscillation by the Third Drive Portion and the Fourth Drive Portion

[0120] The driving of the movable portion **101** by the third driving portion **112a** and the fourth driving portion **112b** will be described. In the third drive portion **112a** and the fourth drive portion **112b**, when a drive voltage is applied in parallel to the piezoelectric portions **202** of the third drive portion **112a** and the fourth drive portion **112b** via the upper electrode **201** and the lower electrode **203**, the piezoelectric portions **202** are deformed. The third drive portion **112a** and the fourth drive portion **112b** are bent and deformed by the deformation of the piezoelectric portion **202**. The third drive portion **112a** and the fourth drive portion **112b** are bent and deformed so that the first torsion bar spring **111a** and the second torsion bar spring **111b** are twisted. A driving force about the first

axis E1 as a rotation axis acts on the movable portion **101** in accordance with the torsion of each of the first torsion bar spring **111a** and the second torsion bar spring **111b**. When this driving force acts, the movable portion **101** oscillates about the first axis E1 as a rotation axis.

[0121] The drive voltages applied to the third drive portion **112a** and the fourth drive portion **112b** are controlled by a control device. For example, a drive voltage having a predetermined sinusoidal waveform is applied to the third drive portion **112a** and the fourth drive portion **112b** in parallel from the control device so that the movable portion **101** can be oscillated about the first axis E1 as a rotation axis at a cycle of the drive voltage having a sinusoidal waveform. When the frequency of the sinusoidal waveform voltage is set to about 20 kilohertz (kHz) that is substantially equal to the resonance frequency of the first torsion bar spring **111a** and the second torsion bar spring **111b**, the movable portion **101** can be resonantly oscillated at about 20 kHz by utilizing the occurrence of mechanical resonance due to the torsion of the first torsion bar spring **111a** and the second torsion bar spring **111b**. The movable portion **101** is resonantly oscillated so that a large displacement of the movable portion **101** can be obtained even when a small drive voltage is applied.

[0122] Oscillation by the Upper Drive Portion and the Lower Drive Portion The oscillation of the movable portion **101** by the upper drive portion **131u** and the lower drive portion **131v** will be described with reference to FIG. 1 and FIGS. 6 to 12. In the movable device **13**, since the first upper drive portion **131u-1**, the second upper drive portion **131u-2**, the first lower drive portion **131v-1**, and the second lower drive portion **131v-2** have the same shape and perform substantially the same function, the oscillation of the movable portion **101** by the second upper drive portion **131u-2** will be described as a representative.

[0123] In order to describe the oscillation by the upper drive portion **131u**, as illustrated in FIG. 1, the second upper drive portion **132a** and the second upper drive portion **132c**, which are odd-numbered from the second upper drive portion **132a** closest to the movable portion **101** among the second upper drive portion **132a**, the second upper drive portion **132b**, the second upper drive portion **132c**, and the second upper drive portion **132d**, are similarly referred to as a drive portion group **200A**. In addition, the second upper drive portion **132b** and the second upper drive portion **132d**, which are even-numbered from the second upper drive portion **132a** closest to the movable portion **101** among the second upper drive portion **132a**, the second upper drive portion **132b**, the second upper drive portion **132c**, and the second upper drive portion **132d**, are similarly referred to as a drive portion group **200B**.

[0124] FIG. 6 is a first schematic perspective view of an example of a state in which the movable portion **101** is oscillated by the second upper drive portion **131u-2** in the movable device **13** according to the first embodiment of the present disclosure. FIG. 7 is a second schematic perspective view of an example of a state in which the movable portion **101** is oscillated by the second upper drive portion **131u-2** in the movable device **13** according to the first embodiment of the present disclosure. FIG. 8 is a third schematic perspective view of an example of a state in which the movable portion **101** is oscillated by the second upper drive portion **131u-2** in the movable device **13** according to the first embodiment of the present disclosure. FIG. 9 is a fourth schematic perspective view of an example of a state in which the movable portion **101** is oscillated by the second upper drive portion **131u-2** in the movable device **13** according to the first embodiment of the present disclosure. FIG. 10 is a first diagram illustrating a waveform of a drive voltage applied to the drive portion group **200A** of the movable device **13** according to the first embodiment of the present disclosure. FIG. 11 is a second diagram illustrating a waveform of a drive voltage applied to the drive portion group **200B** of the movable device **13** according to the first embodiment of the present disclosure. FIG. 12 is a diagram illustrating a waveform in which the waveform of the drive voltage of FIG. 10 and the waveform of the drive voltage of FIG. 11 are superimposed.

[0125] FIGS. 6 to 9 are diagrams illustrating a state where the support portion **120** coupled to the second upper drive portion **131u-2** is oscillated by the second upper drive portion **131u-2**. The

support portion **120** is represented by a broken line. The support portion **120** is oscillated so that the movable portion **101** supported by the support portion **120** is oscillated together with the support portion **120**.

[0126] As illustrated in FIG. 6, in a state in which no drive voltage is applied to the second upper drive portion **131u-2**, the deflection angle of the movable portion **101** due to the support portion **120** oscillated by the second upper drive portion **131u-2** is zero. A virtual plane including the outer edge of the support portion **120** in a top view is substantially parallel to the XY-plane, and the mirror surface **14** of the movable portion **101** supported by the support portion **120** is substantially parallel to the XY-plane.

[0127] As illustrated in FIG. 7, when a drive voltage is applied to the drive portion group **200A** in parallel, the drive portion group **200A** is bent and deformed in the same direction, and the support portion **120** tilts in the negative direction with the second axis E2 as the rotation axis.

[0128] As illustrated in FIG. 8, when the displacement amount of the support portion **120** by the drive portion group **200A** by voltage application and the displacement amount of the support portion **120** by the drive portion group **200B** by voltage application are balanced, the deflection angle is zero.

[0129] As illustrated in FIG. 9, when a drive voltage is applied to the drive portion group **200B** in parallel, the drive portion group **200B** is bent and deformed in the same direction, and the movable portion **101** is oscillated in the +Z-direction about the second axis E2 as the rotation axis.

[0130] A drive voltage is applied to the second upper drive portion **131u-2** so as to continuously repeat the displacement illustrated in FIGS. 7 to 9 so that the support portion **120** can be oscillated about the second axis E2, and the movable portion **101** supported by the support portion **120** can be oscillated around the second axis E2.

[0131] As illustrated in FIGS. 7 and 9, the multiple piezoelectric portions **202** of the drive portion group **200A** are simultaneously bent and deformed or the multiple piezoelectric portions **202** of the drive portion group **200B** are simultaneously bent and deformed, so that the displacement amount due to the bend and deformation is accumulated and the deflection angle of the support portion **120** with the second axis E2 as the rotation axis can be increased. In the movable device **13**, the upper drive portion **131u** and the lower drive portion **131v** are coupled in point symmetry with respect to the center point of the mirror surface **14** that coincides with the center point of the support portion **120**. Thus, when a drive voltage is applied to the drive portion group **200A**, a driving force that moves the movable portion **101** in the +Z-direction is generated at the second end **135-2** of the second upper drive portion **131u-2** coupled to the support portion **120**, and a driving force that moves the movable portion **101** in the -Z-direction is generated at the first end **135-1** of the first upper drive portion **131u-1** coupled to the support portion **120**. The displacement amounts of the first upper drive portion **131u-1** and the second upper drive portion **131u-2** are accumulated, and the deflection angle of the movable portion **101** supported by the support portion **120** with respect to the second axis E2 as the rotation axis can be increased.

[0132] The displacement amount is accumulated in the lower drive portion **131v** as well. In the movable device **13**, the displacement amounts of the upper drive portion **131u** and the lower drive portion **131v** are accumulated, and a larger displacement amount can be obtained. The oscillation of the movable portion **101** by the upper drive beam **112u** and the lower drive beam **112v** is not limited to the non-resonant oscillation, and may be resonant oscillation. Even when the oscillation of the movable portion **101** by the upper drive beam **112u** and the lower drive beam **112v** is non-resonant oscillation, a large displacement amount can be obtained by accumulating the displacement amounts of the upper drive portion **131u** and the lower drive portion **131v**.

[0133] In the movable device **13**, the upper drive beam **112u** is driven by an antiphase drive voltage having a phase opposite to the drive voltage that drives the lower drive beam **112v**. Accordingly, there is no need to arrange an A channel and a B channel in the same substrate as in the meander structure, and the movable device **13** can be formed in a smaller size.

[0134] The drive voltages applied to the upper drive portion **131u** and the lower drive portion **131v** are controlled by the control device. The drive voltages applied to the drive portion group **200A** and the drive portion group **200B** will be described with reference to FIGS. **10** to **12**.

[0135] FIG. **10** is a first diagram illustrating a waveform of a drive voltage applied to the drive portion group **200A** of the movable device **13** according to the first embodiment of the present disclosure. FIG. **11** is a second diagram illustrating a waveform of a drive voltage applied to the drive portion group **200B** of the movable device **13** according to the first embodiment of the present disclosure. FIG. **12** is a diagram illustrating a waveform in which the waveform of the drive voltage illustrated in FIG. **10** and the waveform of the drive voltage illustrated in FIG. **11** are superimposed.

[0136] As illustrated in FIG. **10**, the drive voltage applied to the drive portion group **200A** is, for example, a drive voltage having a sawtooth waveform. The frequency is, for example, 60 hertz (Hz). The waveform of the drive voltage applied to the drive portion group **200A** has a predetermined ratio of, for example, $TrA:TfA=9:1$, where TrA is the time width of the rising period in which the voltage value increases from the minimum value to the next maximum value, and TfA is the time width of the falling period in which the voltage value decreases from the maximum value to the next minimum value. The ratio of TrA to one cycle is referred to as symmetry of the drive voltage applied to the drive portion group **200A**.

[0137] As illustrated in FIG. **11**, the drive voltage applied to the drive portion group **200B** is, for example, a drive voltage having a sawtooth waveform. The frequency is, for example, 60 Hz. The waveform of the drive voltage applied to the drive portion group **200B** has a predetermined ratio of, for example, $TfB:TrB=9:1$, where TrB is the time width of the rising period in which the voltage value increases from the minimum value to the next maximum value, and TfB is the time width of the falling period in which the voltage value decreases from the maximum value to the next minimum value. The ratio of TfB to one cycle is referred to as symmetry of the drive voltage B. As illustrated in FIG. **12**, for example, the period TA of the waveform of the drive voltage applied to the drive portion group **200A** and the period TB of the waveform of the drive voltage applied to the drive portion group **200B** are set to be the same.

[0138] The sawtooth waveforms of the drive voltage applied to the drive portion group **200B** and the drive voltage applied to the drive portion group **200B** are generated by the superposition of sine waves. However, the drive voltage applied to the drive portion group **200B** and the drive voltage applied to the drive portion group **200B** are not limited to the drive voltage having sawtooth waveforms. The waveform may be appropriately changed according to the device characteristics of the movable device **13**, such as a drive voltage having a waveform in which the apex of the sawtooth waveform is rounded, a drive voltage having a waveform in which the straight region of the sawtooth waveform is curved.

Second Embodiment

[0139] A movable device according to a second embodiment of the present disclosure will be described. The same reference numerals and symbols as those of the above-described embodiments denote the same or similar members or structures, and detailed descriptions thereof will be omitted. This point is also applied to other embodiments described below.

[0140] The structure of a movable device according to a second embodiment of the present disclosure will be described with reference to FIGS. **13** to **17**. FIG. **13** is a schematic top view of an example of the movable device **13** according to the second embodiment of the present disclosure. FIG. **14** is a schematic cross-sectional view of the movable device **13** taken along the line XIV-XIV in FIG. **13**. FIG. **15** is a schematic cross-sectional view of the movable device **13** taken along the line XV-XV in FIG. **13**. FIG. **16** is a schematic bottom view of an example of the movable device **13** according to the second embodiment of the present disclosure. FIG. **17** is a schematic cross-sectional view of the movable device **13** taken along the line XVII-XVII in FIG. **13**.

[0141] The movable device **13** according to the second embodiment of the present disclosure

differs from the movable device **13** according to the first embodiment in that the upper drive beam **112u** is electrically connected to the lower drive beam **112v** via a through-silicon via (TSV) **162**. The through-silicon via **162** is an electrode that passes through a hole formed in a silicon substrate to connect upper and lower chips.

[0142] Since the upper drive beam **112u** and the lower drive beam **112v** are electrically connected via the through-silicon via **162**, the wiring between the first SOI substrate **361** and the second SOI substrate **362** that are different substrates can be performed. Since the wiring between the first SOI substrate **361** and the second SOI substrate **362** is performed, a drive voltage can be applied to each of the upper drive beam **112u** and the lower drive beam **112v**.

Third Embodiment

[0143] A movable device according to a third embodiment of the present disclosure will be described with reference to FIGS. **18** to **23**. FIG. **18** is a schematic top view of an example of the movable device **13** according to the third embodiment of the present disclosure. FIG. **19** is a schematic top view of the movable device **13** taken along the line XIX-XIX in FIG. **20**. FIG. **20** is a schematic cross-sectional view of the movable device **13** taken along the line XX-XX in FIG. **18**. FIG. **21** is a schematic cross-sectional view of the movable device **13** taken along the line XXI-XXI in FIG. **18**. FIG. **22** is a schematic bottom view of an example of the movable device **13** according to the third embodiment of the present disclosure. FIG. **23** is a schematic cross-sectional view of the movable device taken along the line XXIII-XXIII in FIG. **18**.

[0144] The movable device **13** according to the third embodiment differs from the movable device **13** according to the first embodiment of the present disclosure in that the shape of the upper drive beam **112u** and the shape of the lower drive beam **112v** are different from each other.

[0145] In the example illustrated in FIG. **19**, among the multiple drive beams of the lower drive beam **112v**, the width of the drive beam on the side closer to the fixed frame **140** is widened to increase the partial rigidity, and the width of the drive beam on the side closer to the support portion **120** is narrowed to decrease the partial rigidity. Accordingly, the resonance frequency can become higher. The higher resonance frequency becomes can stabilize the operation of the movable device **13**.

[0146] The thickness of the silicon active layer **362c** of the second SOI substrate **362** is formed thicker than the thickness of the silicon active layer **361c** of the first SOI substrate **361** so that the thickness of the lower drive beam **112v** may be formed thicker than the thickness of the upper drive beam **112u**. Also, according to this configuration, the resonance frequency can become higher, and the operation of the movable device **13** can be stabilized.

Fourth Embodiment

[0147] A movable device according to a fourth embodiment of the present disclosure will be described with reference to FIGS. **24** to **28**. FIG. **24** is a schematic top view of an example of the movable device **13** according to the fourth embodiment of the present disclosure. FIG. **25** is a schematic cross-sectional view of the movable device **13** taken along the line XXV-XXV in FIG. **24**; FIG. **26** is a schematic cross-sectional view of the movable device **13** taken along the line XXVI-XXVI in FIG. **24**. FIG. **27** is a schematic bottom view of an example of the movable device **13** according to the fourth embodiment of the present disclosure. FIG. **28** is a schematic cross-sectional view of the movable device **13** taken along the line XXVIII-XXVIII in FIG. **24**.

[0148] In the movable device **13** according to the fourth embodiment of the present disclosure, the fifth drive portion **115a** and the sixth drive portion **115b** are disposed above the second SOI substrate **362** so that the displacement amount of the oscillation of the movable portion **101** about the first axis E1 as the rotation axis is increased. Accordingly, the scan range of the light beam incident on the mirror surface **14** in the direction along the second axis E2 is widened. The fifth drive portion **115a** and the sixth drive portion **115b** are examples of elastic support members.

[0149] The third drive portion **112a** and the fourth drive portion **112b** are formed in the silicon active layer **361c** of the first SOI substrate **361**. One end of the third drive portion **112a** is coupled

to the first torsion bar spring **111a**. The other end of the third drive portion **112a** is coupled to the fifth drive portion **115a** via the inter-substrate coupling portion **160**. One end of the fourth drive portion **112b** is coupled to the second torsion bar spring **111b**. The other end of the fourth drive portion **112b** is coupled to the sixth drive portion **115b** via the inter-substrate coupling portion **160**. The other ends of the fifth drive portion **115a** and the sixth drive portion **115b** are coupled to the inner circumferential portion of the support portion **120**.

[0150] The support portion **120** includes a silicon support layer **362a**, a silicon oxide layer **362b**, and a silicon active layer **362c** of the second SOI substrate **362**. The frame of the support portion **120** surrounds the movable portion **101**, the third drive portion **112a**, the fourth drive portion **112b**, the fifth drive portion **115a**, and the sixth drive portion **115b** in a top view.

[0151] In the fourth embodiment of the present disclosure, since the driving force that oscillates the movable portion **101** about the first axis **E1** is obtained from both the third drive portion **112a** and the fifth drive portion **115a**, a large displacement amount can be obtained. Further, the antiphase drive voltage is applied to the third drive portion **112a** and the fifth drive portion **115a** through separate wiring, displacements in both directions are obtained, as in the meander structure in driving the movable portion **101** with the second axis **E2** as a rotation axis, and a large displacement is obtained.

[0152] The lower drive portion **131v** is formed in the silicon active layer **362c** of the second SOI substrate **362**. The lower drive portion **131v** has a meander structure in which multiple drive beams are coupled so as to be folded back. One end of each of the first lower drive beam **112v-1** and the second lower drive beam **112v-2** is coupled to the outer circumferential portion of the support portion **120**. The other end of the first lower drive beam **112v-1** is coupled to the first upper drive beam **112u-1** via the inter-substrate coupling portion **160**. The other end of the second lower drive beam **112v-2** is coupled to the second upper drive beam **112u-2** via the inter-substrate coupling portion **160**.

[0153] The first upper drive beam **112u-1** and the second upper drive beam **112u-2** are formed in the silicon active layer **361c** of the first SOI substrate **361**, and have a meander structure in which multiple drive beams are coupled so as to be folded back.

[0154] One end of the first upper drive beam **112u-1** is coupled to the first lower drive beam **112v-1** via the inter-substrate coupling portion **160**. The other end of the first upper drive beam **112u-1** is coupled to the inner circumferential portion of the fixed frame **140**. In the inter-substrate coupling portion **160**, electrical wiring is connected between the first SOI substrate **361** and the second SOI substrate **362** by the through-silicon via **162** in order to apply a voltage to the piezoelectric portion **202**.

Fifth Embodiment

[0155] A movable device according to a fifth embodiment of the present disclosure will be described with reference to FIGS. **29** to **33**. FIG. **29** is a schematic top view of an example of the movable device **13** according to the fifth embodiment of the present disclosure. FIG. **30** is a schematic cross-sectional view of the movable device **13** taken along the line XXX-XXX in FIG. **29**. FIG. **31** is a schematic cross-sectional view of the movable device **13** taken along the line XXXI-XXXI in FIG. **29**. FIG. **32** is a schematic bottom view of an example of the movable device **13** according to the fifth embodiment of the present disclosure. FIG. **33** is a schematic cross-sectional view of the movable device **13** taken along the line XXXIII-XXXIII in FIG. **29**.

[0156] The movable device **13** according to the fifth embodiment of the present disclosure differs from the movable device **13** according to the first embodiment of the present disclosure in that the second SOI substrate **362** is turned upside down.

[0157] For example, the movable device **13** is formed by bonding two SOI substrates, i.e., a first SOI substrate **361** and a second SOI substrate **362**, with a silicon oxide layer **363** therebetween. The first SOI substrate **361** includes a silicon support layer **361a**, a silicon oxide layer **361b**, and a silicon active layer **361c**, and the second SOI substrate **362** includes a silicon support layer **362a**, a

silicon oxide layer **362b**, and a silicon active layer **362c**.

[0158] In the movable device **13** according to the fifth embodiment of the present disclosure, the silicon support layer **361a** of the first SOI substrate **361** and the silicon support layer **362a** of the second SOI substrate **362** are bonding surfaces. In other words, the second SOI substrate **362** is turned upside down.

[0159] In the movable device **13** according to the fifth embodiment of the present disclosure, the area of the fifth driving portion **115a** can be increased by turning the second SOI substrate **362** upside down, and a larger amount of displacement of the movable portion **101** can be obtained. For example, the lower drive beam **112v** is formed in a shape that avoids the rib **103** so that the fifth drive portion **115a** does not interfere with the rib **103** on the back surface of the movable portion **101** when the lower drive beam **112v** and the fifth drive portion **115a** are driven in the oscillation of the movable portion **101** about the first axis **E1** as the rotation axis. In the fifth embodiment of the present disclosure, the space in the Z-direction obtained due to the silicon support layer **362a** of the second SOI substrate **362** can prevent interference with the rib **103** on the back surface of the movable portion **101**. Accordingly, the area of the fifth drive portion **115a** can be increased, and the displacement of the movable portion **101** in the oscillation about the first axis **E1** as the rotation axis can be increased.

Sixth Embodiment

[0160] A movable device according to a sixth embodiment of the present disclosure will be described with reference to FIGS. **34** to **38**. FIG. **34** is a schematic top view of an example of the movable device **13** according to the sixth embodiment of the present disclosure. FIG. **35** is a schematic cross-sectional view of the movable device **13** taken along the line XXXV-XXXV in FIG. **34**. FIG. **36** is a schematic cross-sectional view of the movable device **13** taken along the line XXXVI-XXXVI in FIG. **34**. FIG. **37** is a schematic bottom view of an example of the movable device **13** according to the sixth embodiment of the present disclosure. FIG. **38** is a schematic cross-sectional view of the movable device **13** taken along the line XXXVIII-XXXVIII in FIG. **34**.

[0161] The movable device **13** according to the sixth embodiment of the present disclosure differs from the movable device **13** according to the first embodiment of the present disclosure in that the size of the movable portion **101** is larger than that of the structure in which the movable portion **101** oscillates about the first axis **E1** as a rotation axis.

[0162] In the movable device **13** according to the sixth embodiment of the present disclosure, the movable portion **101** is formed to be one size larger than the structure in which the movable portion **101** oscillates about the first axis **E1** as the rotation axis, and is larger than the support portion **120**. Since the support portion **120** is formed of a thick silicon support layer, the size of the support portion **120** has a large influence on the moment of inertia. The support portion **120** is formed so as to be smaller than the size of the movable portion **101** so that the moment of inertia can be small.

[0163] In the movable device **13** according to the sixth embodiment of the present disclosure, the upper drive beam **112u** includes four drive beams, and the lower drive beam **112v** includes six drive beams. In other words, the number of drive beams in the upper drive beam **112u** is smaller than the number of drive beams in the lower drive beam **112v**. The number of drive beams in the upper drive beam **112u** is smaller than the number of drive beams in the lower drive beam **112v** so that the movable portion **101** can be larger. In other words, the main frame can be formed relatively smaller.

[0164] In the movable device **13** according to the sixth embodiment of the present disclosure, the rib **103** of the movable portion **101** and the support portion **120** are disposed so as not to overlap each other in the Z-direction. Accordingly, the mutual interference in the deformation in the Z-direction during operation can be prevented.

Embodiment of Application

[0165] Embodiments of various applications to which the movable device **13** according to embodiments of the present disclosure is applied will be described below.

Light Scan System

[0166] First, a light scan system **10** to which the movable device **13** according to an embodiment of the present disclosure is applied will be described in detail with reference to FIGS. **39** to **42**. FIG. **39** is a schematic diagram illustrating an example of the light scan system **10**. As illustrated in FIG. **39**, the light scan system **10** deflects the light beam emitted from a light source device **12** by the mirror surface **14** of the movable device **13** under the control of a control device **11**, and optically scans a scan surface **15** with the light beam.

[0167] The light scan system **10** includes the control device **11**, the light source device **12**, and the movable device **13** having the mirror surface **14**.

[0168] The control device **11** is an electronic circuit unit including, for example, a central processing unit (CPU) and a field-programmable gate array (FPGA). The movable device **13** is, for example, a micro electromechanical systems (MEMS) device having a mirror surface **14** and can move the mirror surface **14**. The light source device **12** is, for example, a laser device to emit a laser beam. The scan surface **15** is, for example, a screen.

[0169] The control device **11** generates a control command for the light source device **12** and the movable device **13** based on the obtained optical scan information, and outputs a drive signal to the light source device **12** and the movable device **13** based on the control command.

[0170] The light source device **12** irradiates an object with a light beam based on the input drive signal. The movable device **13** moves the mirror surface **14** in at least one of one-axial direction or two-axial directions based on the input drive signal.

[0171] Accordingly, for example, the mirror surface **14** of the movable device **13** is reciprocated in two-axial directions within a predetermined range by the control device **11** based on image information as an example of optical scan information so that the irradiation light beam from the light source device **12** that goes in the mirror surface **14** is deflected around a certain axis to perform optical scan. As a result, any image is projected onto the scan surface **15**.

[0172] A hardware configuration of an example of the light scan system **10** will be described with reference to FIG. **40**. FIG. **40** is a diagram illustrating a hardware configuration of a light scan system **10**. As illustrated in FIG. **40**, the light scan system **10** includes a control device **11**, a light source device **12**, and a movable device **13**, which are electrically connected. The control device **11** includes a CPU **20**, a random-access memory (RAM) **21**, a read-only memory (ROM) **22**, an FPGA **23**, an external input and output (I/F) **24**, a light source driver **25**, and a movable device driver **26**.

[0173] The CPU **20** is a computing device that reads programs and data from a storage device such as the ROM **22** to the RAM **21** and executes processing to implement the overall control and functions of the control device **11**. The RAM **21** is a volatile storage device to temporarily hold programs and data. The ROM **22** is a nonvolatile storage device that can retain programs and data even when the power is turned off, and stores processing programs and data executed by the CPU **20** to control the functions of the light scan system **10**. The FPGA **23** is a circuit that outputs control signals suitable for the light source driver **25** and the movable device driver **26** in accordance with the processing of the CPU **20**.

[0174] The external I/F **24** is, for example, an interface with an external device or a network. Examples of the external device include a host device such as a personal computer (PC), and a storage device such as a universal serial bus (USB) memory, an SD card, a compact disk (CD), a digital versatile disk (DVD), a hard disk drive (HDD), or a solid-state drive (SSD). The network is, for example, a controller area network (CAN) for an automobile, a local area network (LAN), and the Internet. The external I/F **24** may enable connection or communication with the external device, and the external I/F **24** may be prepared for each external device.

[0175] The light source driver is an electric circuit that outputs a drive signal such as a drive voltage to the light source device **12** in accordance with an input control signal. The movable device driver **26** is an electric circuit that outputs a drive signal such as a drive voltage to the movable device **13** in accordance with an input control signal. In the control device **11**, the CPU **20**

acquires optical scan information from an external device or a network via the external I/F **24**. Any configuration is acceptable as long as the CPU **20** can acquire the optical scan information, and the optical scan information may be stored in the ROM **22** or the FPGA **23** in the control device **11**. Alternatively, for example, a storage device such as an SSD may be additionally disposed in the control device **11** and the optical scan information may be stored in the storage device.

[0176] The optical scan information is information indicating how the scan surface **15** is optically scanned, and for example, when an image is displayed by optical scan, the optical scan information is image data. In addition, for example, when optical writing is performed by optical scan, the optical scan information is writing data indicating a writing order and a writing position. Further, for example, when an object is recognized by optical scan, the optical scan information includes irradiation data indicating the irradiation timing and an irradiation range with a light beam for object recognition.

[0177] The control device **11** can implement a functional configuration described below by an instruction of the CPU **20** and the hardware configuration illustrated in FIG. **40**.

[0178] The functional configuration of the control device **11** of the light scan system **10** will be described with reference to FIG. **41**. FIG. **41** is a functional block diagram illustrating an example of a control device of the light scan system. As illustrated in FIG. **41**, the control device **11** includes a control unit **30** and a drive signal output unit **31** as functions.

[0179] The control unit **30** is implemented by, for example, the CPU **20** or the FPGA **23**, obtains the optical scan information from an external device, converts the optical scan information into a control signal, and outputs the control signal to the drive signal output unit **31**. For example, the control unit **30** obtains image data as optical scan information from, for example, an external device, generates a control signal from the image data by predetermined processing, and output the control signal to the drive signal output unit **31**. The drive signal output unit **31** is implemented by the light source driver **25** and the movable device driver **26**, and outputs a drive signal to the light source device **12** or the movable device **13** based on the input control signal.

[0180] The drive signal is a signal for controlling the drive of the light source device **12** or the movable device **13**. For example, in the light source device **12**, the drive voltage is used to control the irradiation timing and the irradiation intensity of the light source. In addition, for example, in the movable device **13**, the drive voltage is a drive voltage for controlling the timing and the movable range of the mirror surface **14** of the movable device **13**.

[0181] The process in which the light scan system **10** optically scans the scan surface **15** with a light beam will be described with reference to FIG. **42**. FIG. **42** is a flowchart of an example of a process related to an optical scanner.

[0182] In step **S11**, the control unit **30** obtains optical scan information from an external device. In step **S12**, the control unit **30** generates a control signal from the obtained optical scan information and outputs the control signal to the drive signal output unit **31**. In step **S13**, the drive signal output unit **31** outputs a drive signal to the light source device **12** and the movable device **13** based on the input control signal.

[0183] In step **S14**, the light source device **12** performs light irradiation based on the input drive signal. The movable device **13** moves the mirror surface **14** based on the input drive signal. The light source device **12** and the movable device **13** are driven so that the light beam is deflected in any direction, and optically scan is performed.

[0184] In the light scan system **10** described above, a single control device **11** includes a device and a function of controlling the light source device **12** and the movable device **13**, but the device and the function may be disposed separately from the control device for the light source device and the control device for the movable device.

[0185] In the light scan system **10**, the functions of the control unit **30** of the light source device **12** and the movable device **13** and the function of the drive signal output unit **31** are disposed in one control device **11**, but these functions may be disposed separately. For example, a drive signal

output device including the drive signal output unit **31** is disposed separately from the control device **11** including the control unit **30**. The light scan system **10** described above may include an optical deflection system to perform light deflection by the movable device **13** having the mirror surface **14**, and the single control device **11**.

[0186] The light scan system **10** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability. As a result, the light scan system **10** can scan a wide range with a light beam with high accuracy.

Image Projection Apparatus

[0187] An image projection apparatus including the movable device according to the present embodiment will be described in detail with reference to FIGS. **43** and **44**. FIG. **43** is a schematic view of an automobile **400** including a head-up display device **500** as an example of an image projection apparatus. FIG. **44** is a schematic diagram illustrating an example of a head-up display device **500**.

[0188] The image projection apparatus is an apparatus that projects an image by optical scan, and is, for example, a head-up display device.

[0189] As illustrated in FIG. **43**, the head-up display device **500** is installed, for example, near a windshield (e.g., a front window **401**) of an automobile **400**. The projection light beam L emitted from the head-up display device **500** is reflected by the front window **401** and directed to an observer (e.g., a driver **402**) that is a user. Accordingly, the driver **402** can visually recognize the image projected by the head-up display device **500** as a virtual image. A combiner may be disposed on an inner wall surface of the windshield, and the user may visually recognize a virtual image by the projected light beam reflected by the combiner.

[0190] As illustrated in FIG. **44**, in the head-up display device **500**, a red laser light source **501R** emits a red laser beam, a green laser light source **501G** emits a green laser beam, and a blue laser light source **501B** emits a blue laser beam. These three emitted laser beams pass through an incident optical system, and are deflected by a movable device **13** including a mirror surface **14**. The incident optical system includes a collimator lens **502**, a collimator lens **503**, and a collimator lens **504**, which are disposed for the red laser beam, the green laser beam, and the blue laser beam, respectively, two dichroic mirrors **505** and **506**, and a light amount adjusting unit **507**. The deflected laser beams are projected onto a screen through a projection optical system including a free-form surface mirror **509**, an intermediate screen **510**, and a projection mirror **511**. In the head-up display device **500**, the red laser light source **501R**, the green laser light source **501G**, and the blue laser light source **501B**, the collimator lens **502**, the collimator lens **503**, the collimator lens **504**, the dichroic mirror **505**, and the dichroic mirror **506** are formed as a light source unit **530** by an optical housing.

[0191] The head-up display device **500** described above projects an intermediate image displayed on the intermediate screen **510** on the front window **401** of the automobile **400** so that the driver **402** can be visually recognized the intermediate image as a virtual image.

[0192] The red laser beam emitted from the red laser light source **501R**, the green laser beam emitted from the green laser light source **501G**, and the blue laser beam emitted from the blue laser light source **501B** are collimated by the collimator lens **502**, the collimator lens **503**, and the collimator lens **504**, respectively, as substantially parallel laser beams, and are combined by the two dichroic mirrors **505** and **506**. The amount of the combined laser beam is adjusted by the light amount adjusting unit **507**, and the movable device **13** including the mirror surface **14** two-dimensionally scans the free-form surface mirror **509** with the laser beam. The projection light beam L with which the movable device **13** two-dimensionally scans the free-form surface mirror **509** is reflected by the free-form surface mirror **509** is corrected in distortion, and is condensed on the intermediate screen **510** to display an intermediate image. The intermediate screen **510** includes a microlens array in which microlenses are two-dimensionally arranged, and enlarges the projection light beam L incident on the intermediate screen **510** at each microlens.

[0193] The movable device **13** reciprocates the mirror surface **14** in two-axial directions and two-dimensionally scans the free-form surface mirror **509** with the projection light beam. The control of the movable device **13** is performed in synchronism with the timing of light emission of the red laser light source **501R**, the green laser light source **501G**, and the blue laser light source **501B**.
[0194] As described above, the head-up display device **500** has been described as an example of the image projection apparatus, but the image projection apparatus may be any device that projects an image by performing optical scan with the movable device **13** including the mirror surface **14**. For example, the image projection apparatus can be applied to a projector that is placed on a desk and projects an image on a display screen, and a head-mounted display device that projects an image to a reflective-and-transmissive screen included in a mounting member mounted on an observer's head or projects an image to an eyeball as a screen.

[0195] The image projection apparatus may be included in not only a vehicle and a mounting member but also, for example, a moving body such as an aircraft, a ship, and a mobile robot, or a non-moving body such as a work robot that operates a drive object such as a manipulator without moving from the site.

[0196] The head-up display device **500** is an example of a head-up display. The automobile **400** is an example of a vehicle.

[0197] Since the head-up display device **500** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, a large screen image with high image quality can be displayed.

Optical Writing Apparatus

[0198] An optical writing apparatus to which the movable device **13** of the present embodiment is applied will be described in detail with reference to FIGS. **45** and **46**.

[0199] FIG. **45** is a schematic diagram illustrating an example of an image forming apparatus including an optical writing apparatus **600**. FIG. **46** is a schematic diagram illustrating an example of the optical writing apparatus **600**.

[0200] As illustrated in FIG. **45**, the optical writing apparatus **600** is used as a unit of an image forming apparatus such as a laser printer **650** having a printer function using a laser beam. In the image forming apparatus, the optical writing apparatus **600** optically scans a photoconductive drum that serves as a scan surface **15** with one or multiple laser beams, to perform optical writing to the photoconductive drum.

[0201] As illustrated in FIG. **46**, in the optical writing apparatus **600**, a laser beam from a light source device **12** such as a laser element passes through an imaging optical system **601** such as a collimator lens, and is deflected in one-axial direction or two-axial directions by a movable device **13** including a mirror surface **14**. The laser beam deflected by the movable device **13** passes through the scanning optical system **602** including a first lens **602a**, a second lens **602b**, and a reflecting mirror portion **602c**, and the optical writing apparatus **600** irradiates the scan surface **15** (e.g., photoconductive drum or photosensitive paper) with the laser beam to perform optical writing. The scanning optical system **602** forms an image of a light beam in a spot shape on the scan surface **15**. The light source device **12** and the movable device **13** including the mirror surface **14** are driven based on the control of the control device **11**.

[0202] As described above, the optical writing apparatus **600** can be used as a member of an image forming apparatus having a printer function by a laser beam. Further, the optical writing apparatus **600** can change the scanning optical system to perform optical scan not only in one-axial direction but also two-axial directions, and thus can be used as a unit of the image forming apparatus such as a laser label apparatus that deflects the laser beam to a thermal medium to print by heating.

[0203] Since the movable device **13** including the mirror surface **14** applied to the optical writing apparatus has a smaller power consumption for driving than a rotary polygon mirror using a polygon mirror, and the movable device **13** has an advantage in saving power of the optical writing apparatus. In addition, since the wind noise generated when the movable device **13** is vibrated is

smaller than the wind noise of the rotary polygon mirror, it is advantageous for increasing the quietness of the optical writing apparatus. The optical writing apparatus requires a much smaller installation space than the rotary polygon mirror, and the movable device **13** generates a small amount of heat, so that the optical writing apparatus can be easily reduced in size. As a result, the image forming apparatus has an advantage in reducing in size.

[0204] Since the optical writing apparatus **600** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, an image with high image quality can be formed on a large region of a recording medium.

Object Recognition Apparatus

[0205] An object recognition apparatus to which the movable device of the present embodiment is applied will be described in detail with reference to FIGS. **47** and **48**.

[0206] FIG. **47** is a schematic diagram illustrating an automobile in which a laser imaging detection and ranging (LiDAR) apparatus that is an example of an object recognition apparatus is mounted. In the schematic diagram, the LiDAR apparatus is mounted on a lighting unit including a headlight of the automobile. FIG. **48** is a schematic diagram illustrating an example of a LiDAR apparatus.

[0207] The object recognition apparatus is an apparatus that recognizes an object in a target direction, and is, for example, a LiDAR apparatus. As illustrated in FIG. **47**, the LiDAR apparatus **700** is mounted on, for example, an automobile **701**, and scans a region with a light beam in a predetermined direction, and recognizes the object **702** in the region by receiving the reflected light beam from the object **702**.

[0208] As illustrated in FIG. **48**, the movable device **13** includes an incident optical system including a collimate lens **703** that is an optical system to form a divergent light beam into a substantially parallel light beam, and a plane mirror **704**. The movable device **13** including the mirror surface **14** scans an object with the laser beam that is emitted from the light source device **12** and passes through the incident optical system in a one-axial direction or two-axial directions. The substantially parallel light deflected by the movable device **13** reaches the object **702** in front of the LiDAR apparatus **700** via a projection lens **705** that is a projection optical system. The control device **11** drives the light source device **12** and the movable device **13**. The light detector **709** detects the reflected light beam reflected by the object **702**. In other words, the reflected light beam passes through a condenser lens **706** that is an optical system for receiving and detecting the incident light beam, and is received by an image sensor **707**. The image sensor **707** outputs a detected signal to a signal processing circuit **708**. The signal processing circuit **708** performs predetermined processing such as binarization or noise processing on the received detection signal, and outputs the result to a distance measurement circuit **710**.

[0209] The distance measurement circuit **710** determines the presence or absence of the object **702** according to a time difference between a timing at which the light source device **12** emits the laser beam and a timing at which the light detector **709** receives the laser beam, or a phase difference for each pixel of the image sensor **707** that receives the laser beam, and calculates distance information with respect to the object **702**.

[0210] Since the movable device **13** including the mirror surface **14** is less likely to be damaged as compared with a polygon mirror and is smaller in size, a smaller LiDAR apparatus having higher durability can be provided. When the LiDAR apparatus is attached to, for example, a vehicle, an aircraft, a ship, or a robot, the presence or absence of an obstacle or an object can be determined or the distance to the obstacle or the object can be recognized by performing optical scan within a predetermined range.

[0211] In the object recognition apparatus described above, the LiDAR apparatus **700** is described as an example. However, the object recognition apparatus is not limited to the embodiment described above as long as the object recognition apparatus performs optical scan by the movable device **13** including the mirror surface **14** under control of the control device **11** and receives the

reflection light beam by the light detector to recognize the object **702**.

[0212] The object recognition apparatus described above can be applied to, for example, a biometric authentication apparatus that calculates object information such as a shape from the distance information obtained by scanning a hand or a face and recognizes the object by referring the record, a security sensor that recognizes an intruder by optically scanning the object region, or a unit of the three-dimensional scanner that outputs three-dimensional data.

[0213] Since the object recognition apparatus includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, the object recognition apparatus can recognize an object in a wide range with high accuracy.

Laser Headlamp

[0214] A laser headlamp **50** in which the movable device of the present embodiment is applied to a headlight of an automobile will be described with reference to FIG. **49**. FIG. **49** is a schematic diagram illustrating an example of a configuration of the laser headlamp **50**.

[0215] The laser headlamp **50** includes a control device **11**, a light source device **12b**, a movable device **13** including a mirror surface **14**, a mirror **51**, and a transparent plate **52**.

[0216] The light source device **12b** is a light source that emits a blue laser beam. The light beam emitted from the light source device **12b** goes in the movable device **13** and is reflected by the mirror surface **14**. The movable device **13** moves the mirror surface in the X- and Y-directions based on the signal from the control device **11**, and two-dimensionally scans an object with the blue laser beam from the light source device **12b** in the X- and Y-directions.

[0217] The scan light beam by the movable device **13** is reflected by the mirror **51** and goes in the transparent plate **52**. The front surface or the back surface of the transparent plate **52** is coated with a yellow fluorescent material. The blue laser beam from the mirror **51** changes to white in the legalized range of the color of the headlight when passing through the yellow phosphor coating of the transparent plate **52**. Thus, the area ahead of the automobile is illuminated with the white light beam from the transparent plate **52**.

[0218] The scan light beam by the movable device **13** is scattered in a predetermined manner when passing through the fluorescent material of the transparent plate **52**. Accordingly, the glare of the illumination object ahead of the automobile is reduced.

[0219] When the movable device **13** is applied to a headlight of an automobile, the colors of the light source device **12b** and the fluorescent material are not limited to blue and yellow, respectively. For example, the light source device **12b** may be a near-ultraviolet light source, and the transparent plate **52** may be coated with a mixture of uniformly mixed phosphors of blue, green, and red, which are the three primary colors of light. In this case, the light beam passing through the transparent plate **52** can be converted into white light, and the area ahead of the automobile can be illuminated with white light beam.

[0220] Since the laser headlamp **50** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, the laser headlamp can irradiate a wide region with a laser beam with high accuracy.

Head-Mounted Display

[0221] A head-mounted display (HMD) **60** to which the movable device of the present embodiment is applied will be described with reference to FIGS. **50** and **51**. The HMD **60** is a head-mounted display that can be mounted on the human head, and has a shape of, for example, glasses.

[0222] FIG. **50** is a perspective view of an external appearance of the HMD **60**. In FIG. **50**, the HMD **60** includes fronts **60a** symmetrically disposed in a pair on the left and right sides and temples **60b**. The fronts **60a** can be formed by, for example, light guide plates **61**, and an optical system and a control device can be built in the temples **60b**.

[0223] FIG. **51** is a diagram illustrating a partial configuration of the HMD **60**. Although FIG. **51** illustrates the partial configuration for the left eye, the HMD **60** has the same configuration for the right eye.

[0224] The HMD **60** includes a control device **11**, a light source unit **530**, a light amount adjusting unit **507**, a movable device **13** including a mirror surface **14**, a light guide plate **61**, and a half mirror **62**.

[0225] As described above, the light source unit **530** includes the red laser light source **501R**, the green laser light source **501G**, the blue laser light source **501B**, the collimator lens **502**, the collimator lens **503**, the collimator lens **504**, the dichroic mirror **505**, and the dichroic mirror **506** in an optical housing as a unit. In the light source unit **530**, the red laser beam from the red laser light source **501R**, the green laser beam from the green laser light source **501G**, and the blue laser beam from the blue laser light source **501B** are combined by the dichroic mirror **505** and the dichroic mirror **506**. The light source unit **530** emits the combined parallel light beam.

[0226] The light beam from the light source unit **530** is adjusted by the light amount adjusting unit **507** in its light amount and goes in the movable device **13**. The movable device **13** moves the mirror surface **14** in the X- and Y-directions based on the signal from the control device **11**, and two-dimensionally scans an object with the laser beam from the light source unit **530**. The control of the movable device **13** is performed in synchronism with the timing of light emission of the red laser light source **501R**, the green laser light source **501G**, and the blue laser light source **501B**, and a color image is formed by optical scan.

[0227] The scan light beam by the movable device **13** goes in the light guide plate **61**. The light guide plate **61** guides the scan light beam to the half mirror **62** while reflecting the scan light beam on the inner wall surfaces. The light guide plate **61** is made of a resin having a transparency with respect to the wavelength of the scan light beam.

[0228] The half mirror **62** reflects the light beam from the light guide plate **61** to the back surface side of the HMD **60** and emits the light beam in the direction of the eye of a wearer **63** of the HMD **60**. The half mirror **62** has, for example, a free-form surface shape. The image by the scan light beam is reflected by the half mirror **62**, and is imaged on the retina of the wearer **63**. Alternatively, the image is formed on the retina of the wearer **63** by reflection at the half mirror **62** and the lens effect of the lens of the eyeball. The spatial distortion of the image is corrected by the reflection at the half mirror **62**. The wearer **63** can observe the image formed with the light beam scanned in the X- and Y-directions.

[0229] Since the half mirror **62** is used, the wearer **63** observes a superimposed image of an image of a light beam from the outside and an image by scan light beam. A mirror may be used instead of the half mirror **62** so that the light beam from the outside is eliminated, and only an image by the scan light beam can be observed.

[0230] Since the HMD **60** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, the HMD **60** can display a large screen image with high image quality.

[0231] Tilt-Position Detection Apparatus of Eyeball (Position Detection Apparatus of Pupil or Cornea)

[0232] A tilt-position detection apparatus of an eyeball that includes the movable device **13** will be below. The tilt-position detection apparatus for the eyeball is a pupil-or-cornea position detection apparatus **80**. FIG. **52** is a first schematic block diagram illustrating an example of a pupil-or-cornea position detection apparatus **80**. FIG. **53** is a second schematic block diagram illustrating an example of the pupil-or-cornea position detection apparatus **80**.

[0233] The “tilt-position of the eyeball” in the present embodiment is the position of the pupil-or-cornea of the eyeball or the direction of the user's line of sight. In the following description, the “tilt-position of the eyeball” is the position of the pupil or the cornea, and the “tilt-position detection apparatus of the eyeball” is the “pupil-or-cornea position detection apparatus.” The pupil-or-cornea position detection apparatus described below is the same as a line-of-sight direction tracking apparatus (eye tracking apparatus) that detects or tracks the direction of the user's line of sight continuously or at intervals of time.

[0234] The pupil-or-cornea position detection apparatus **80** illustrated in FIG. 52 includes a light source **82**, a first light deflection part **83**, a movable device **13**, a second light deflection part **85**, and a light receiving unit **86**.

[0235] The light source **82** includes, for example, a red laser light source **82r** to emit a red laser beam, a green laser light source **82g** to emit a green laser beam, a blue laser light source **82b** to a blue laser light beam, and an infrared laser light source **82ir** to emit an infrared laser beam. The red laser light sources **82r**, the green laser light source **82g**, and the blue laser light source **82b** may be any one or a combination of two. The red laser light sources **82r**, the green laser light source **82g**, and the blue laser light source **82b** emit laser beams to form an image by the movable device **13**.

[0236] The infrared laser light source **82ir** emits a light beam to detect the position of the pupil or the cornea. The light beam for detecting the position of the pupil or the cornea is not limited to an infrared light beam, and may be a visible light beam. The light beam for detecting the position of the pupil or the cornea is preferably an invisible light beam from the viewpoint of increasing the visibility of a drawn image.

[0237] The first light deflection part **83** is, for example, a dichroic mirror, and deflects the light beam emitted from the light source **82** toward the mirror surface **14** of the movable device **13** while combining the light beams. The pupil-or-cornea position detection apparatus **80** may include multiple first light deflection parts **83-1**, **83-2**, **83-3**, **83-4**, and **83-5**, in accordance with the number of the laser light sources (i.e., the red laser light sources **82r**, the green laser light source **82g**, the blue laser light source **82b**, and the infrared laser light source **82ir**). The first light deflection part **83** includes multiple first light deflection parts **83-1**, **83-2**, **83-3**, **83-4**, and **83-5**. The multiple first light deflection parts **83-1**, **83-2**, **83-3**, **83-4**, and **83-5** deflect the light beams while combining the light beams.

[0238] The movable device **13** includes a mirror surface **14** and scans the second light deflection part **85** with the light beam deflected by the first light deflection part **83** in two-dimensional directions. At this time, the movable device **13** scans the second light deflection part **85** with the light beam deflected by the first light deflection part **83** using, for example, raster scan to form an image. The movable device **13** can scan the second light deflection part **85** with the light beam deflected by the first light deflecting part **83** by spiral scan.

[0239] The second light deflection part **85** is, for example, a holographic optical element, and deflects the light beam **L1** with which the movable device **13** scanned toward the eyeball **87** of the user. At least a portion of the light beam **L2** deflected by the second light deflection part **85** goes in the eyeball **87** of the user as display image light beam. The second light deflection part **85** may include multiple light deflection members. For example, multiple light deflection parts that reflect specific light beams among the light beams emitted from the light source **82** may be used, and the mirror surface may be different for each light beams emitted from the light source **82**. As a specific example, a structure in which a light deflection part that reflects light beams emitted from the red laser light source **82r**, the green laser light source **82g**, and the blue laser light source **82b** and a light deflection member for reflecting the light beam emitted from the infrared laser light source **82ir** are laminated in order of proximity to the eyeball **87** can be used.

[0240] The light receiving unit **86** receives the light beam **L3** reflected by the eyeball **87** of the user among the light beam **L2** deflected by the second light deflection part **85**, and outputs a detection signal **SD** corresponding to the received light beam. The light receiving unit **86** is an image sensor that can detect, for example, an infrared light beam. The light receiving unit **86** may include multiple light receiving units and be disposed at positions that can receive the light beam **L3** reflected by the eyeball **87** of the user. The light beam received by the light receiving unit **86** changes in intensity depending on the change in position of the eyeball (e.g., pupil or cornea), that is, the change in the direction of the line of sight. Thus, the pupil-or-cornea position detection apparatus **80** of the pupil or the cornea in the present embodiment detects or estimates the position of the pupil or the cornea based on the intensity of the light beam received by the light receiving

unit **86**. The light receiving unit **86** may be configured to image the eyeball **87** irradiated with the light beam L2 deflected by the second light deflection part **85**. In this case, the pupil-or-cornea position detection apparatus **80** detects or estimates the tilt-position of the eyeball **87** based on the position of the pupil or cornea included in the captured image (i.e., detection signal SD) and the position at which the light beam L2 deflected by the second light deflection part **85** is reflected in the eyeball **87**.

[0241] As described above, the pupil-or-cornea position detection apparatus **80** according to the present embodiment can detect the position of the pupil or cornea while forming an image by the movable device **13**. Further, since the movable device **13** has a configuration that can perform optical scan more efficiently, image formation and detection of the position of the pupil or cornea can be implemented with lower power. Further, the movable device **13** can obtain the above-described effect without changing the area necessary for mounting the pupil-or-cornea position detection apparatus **80** as compared with the configuration in the related art. Accordingly, a configuration in which the size of the pupil-or-cornea position detection apparatus **80** is not increased can be implemented.

[0242] The pupil-or-cornea position detection apparatus **80** can be mounted on a head-mounted display as, for example, an eye tracking apparatus, and can detect or track the direction of the user's line of sight. In this case, for example, the resolution of an image displayed in a region near the direction of the user's line of sight in other regions is reduced (i.e., foveal rendering) so that the image processing can be speeded up as compared with the case where a high-resolution image is displayed in the entire region.

[0243] As described above, preferable embodiments have been described in detail. However, embodiments according to the present disclosure are not limited to the embodiments described above, and various modifications and substitutions can be made to the above-described embodiments of the present disclosure without departing from the scope of the claims.

[0244] All the numerals such as ordinal numbers and numbers and reference symbols used in the description of the embodiments are illustrative for specifically describing the technique of the present invention, and the present invention is not limited to the illustrated numerals. In addition, a coupling relation between the components is an example for specifically describing the technology of the present disclosure, and a connection relation for implementing a function of the present disclosure is not limited thereto.

[0245] Aspects of the present disclosure are as follows, for example.

First Aspect

[0246] A movable device includes a movable portion, a drive beam whose one end is directly or indirectly coupled to the movable portion, and a fixed frame to which another end of the drive beam is coupled. The drive beam includes multiple drive beams each including a drive portion. Each of the multiple drive beams includes an upper drive beam and a lower drive beam disposed in a normal direction of the fixed frame so as to be separated from the upper drive beam.

Second Aspect

[0247] In the movable device according to the first aspect, the upper drive beam overlaps the lower drive beam as viewed from the normal direction of the fixed frame.

Third Aspect

[0248] In the movable device according to the first or second aspect, the upper drive beam and the lower drive beam are different in number.

Fourth Aspect

[0249] In the movable device according to any one of the first to third aspects, the upper drive beam is coupled to the lower drive beam by an adhesive.

Fifth Aspect

[0250] In the movable device according to any one of the first to fourth aspects, the upper drive beam is driven by an antiphase drive voltage having a phase opposite to a phase of the drive

voltage for driving the lower drive beam.

Sixth Aspect

[0251] The movable device according to any one of the first to fifth aspects further includes an elastic support member disposed between the movable portion and one of the upper drive beam and the lower drive beam. The movable portion is resonantly driven.

Seventh Aspect

[0252] In the movable device according to any one of the first to sixth aspects, the movable portion is arranged so as to be spaced from either the upper drive beam or the lower drive beam in a normal direction of the fixed frame.

Eighth Aspect

[0253] In the movable device according to any one of the first to seventh aspects, the upper drive beam is electrically connected to the lower drive beam by wire bonding.

Ninth Aspect

[0254] In the movable device according to any one of the first to seventh aspects, the upper drive beam is electrically connected to the lower drive beam via a through-silicon via.

Tenth Aspect

[0255] An image projection apparatus includes the movable device according to any one of the first to ninth aspects.

Eleventh Aspect

[0256] A head-up display includes the movable device according to any one of the first to ninth aspects.

Twelfth Aspect

[0257] A laser headlamp includes the movable device according to any one of the first to ninth aspects.

Thirteenth Aspect

[0258] A head-mounted display includes the movable device according to any one of the first to ninth aspects.

Fourteenth Aspect

[0259] An object recognition apparatus includes the movable device according to any one of the first to ninth aspects.

Fifteenth Aspect

[0260] A moving body includes at least one of the head-up display according to the eleventh aspect, the laser headlamp according to the twelfth aspect, or the object recognition apparatus according to the fourteenth aspect.

[0261] The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose processors, integrated circuits, ASICs (“Application Specific Integrated Circuits”), FPGAs (“Field-Programmable Gate Arrays”), and/or combinations thereof which are configured or programmed, using one or more programs stored in one or more memories, to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carry out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein which is programmed or configured to carry out the recited functionality.

Claims

1. A movable device comprising: a fixed frame; a movable portion; multiple drive beams to drive the movable portion, each of the multiple drive beams having: an end directly or indirectly coupled to the movable portion; and another end coupled to the fixed frame, and each of the multiple drive beams including: an upper drive beam; and a lower drive beam separated from the upper drive

beam in a normal direction of the fixed frame.

2. The movable device according to claim 1, wherein the upper drive beam overlaps the lower drive beam in the normal direction of the fixed frame.
 3. The movable device according to claim 1, wherein the multiple drive beams include: multiple upper drive beams including a first number of the upper drive beam; and multiple lower drive beams including a second number different from the first number of the lower drive beam.
 4. The movable device according to claim 1, wherein the upper drive beam is coupled to the lower drive beam with an adhesive.
 5. The movable device according to claim 1, wherein the upper drive beam is driven by a first drive voltage; and the lower drive beam is driven by a second drive voltage antiphase to the first drive voltage.
 6. The movable device according to claim 1, further comprising: an elastic support portion disposed between the movable portion and one of the upper drive beam or the lower drive beam, wherein the movable portion is resonantly driven by one of the upper drive beam or the lower drive beam via the elastic support portion.
 7. The movable device according to claim 1, wherein the movable portion is separated from one of the upper drive beam or the lower drive beam in the normal direction of the fixed frame.
 8. The movable device according to claim 1, further comprising: a wire bonding electrically connecting the upper drive beam and the lower drive beam.
 9. The movable device according to claim 1, further comprising: a through-silicon via electrically connecting the upper drive beam and the lower drive beam.
 10. An image projection apparatus comprising: the movable device according to claim 1.
 11. A head-up display comprising: the movable device according to claim 1.
 12. A laser headlamp comprising: the movable device according to claim 1.
 13. A head-mounted display comprising: the movable device according to claim 1.
 14. An object recognition apparatus comprising: the movable device according to claim 1.
 15. A moving body comprising: the head-up display according to claim 11.
 16. A moving body comprising: the laser headlamp according to claim 12.
 17. A moving body comprising: the object recognition apparatus according to claim 14.
 18. The movable device according to claim 1, wherein the movable portion is in the fixed frame, and each of the multiple drive beams is between the movable portion and the fixed frame in a direction orthogonal to the normal direction.
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